

Crypto Market Metrics Automation

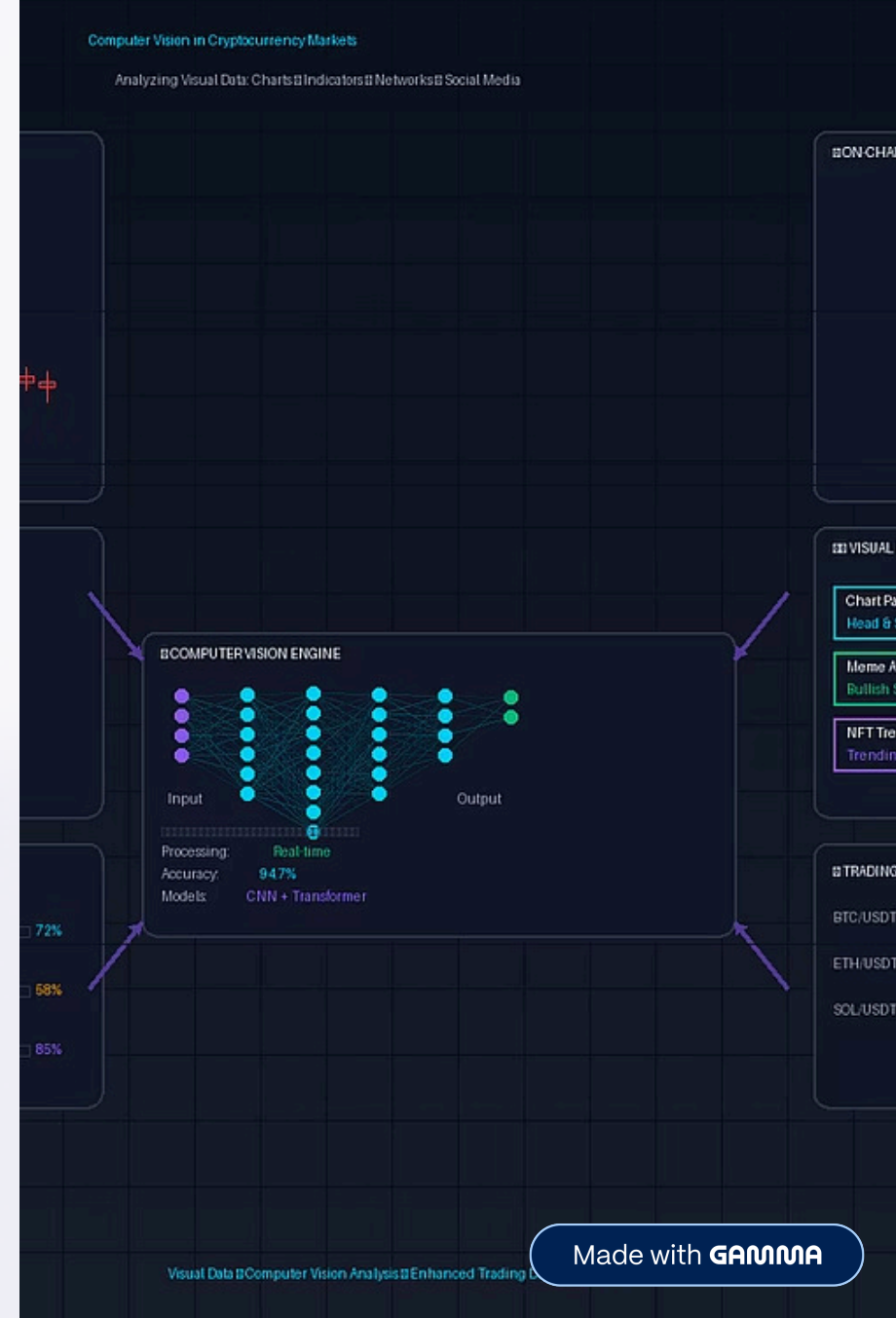
End-to-End Analytics Pipeline using CoinGecko API, Python, Neon PostgreSQL, and Apache Superset

Deepak Nepram

PGDM — Business Analytics / Data Science

IIBS Bangalore · 2026 Portfolio Project

 An automated crypto market intelligence pipeline for measuring price behavior, risk, momentum, and trend performance across major digital assets.



Problem Statement and Objective



The Challenge

- Crypto markets are **highly volatile**, data-rich, and impossible to monitor manually
- Investors need automated tracking of returns, volatility, drawdowns, and momentum
- Raw API data alone is **not decision-ready**
- A repeatable pipeline is required to transform data into actionable business intelligence

❏ **Objective:** Build an end-to-end analytics workflow that automatically fetches crypto market data, computes risk-return metrics, stores structured outputs in PostgreSQL, and visualizes results through an interactive Apache Superset dashboard.

⚡ **Faster Analysis**

📊 **Centralized Metrics**

📈 **Better Decisions**

System Architecture



Pipeline covers **BTC, ETH, and SOL** over a **365-day historical window**, producing two core tables: `crypto_timeseries` and `crypto_snapshot`.

Pipeline and Metrics Engine

- 1

Fetch
Pull daily market chart data from CoinGecko API
- 2

Transform
Convert API response into tidy pandas DataFrames
- 3

Compute
Calculate analytical metrics per coin
- 4

Store
Push final tables into Neon PostgreSQL

Metric	Business Meaning
Daily Return	Day-over-day % price change
Log Return	Continuous compounding return
30-Day Ann. Volatility	Annualized risk level
50-Day Moving Avg	Short-term trend signal
200-Day Moving Avg	Long-term trend baseline
Drawdown %	Loss from peak price
Sharpe Ratio Approx.	Risk-adjusted return proxy
7-Day Momentum	Short-term directional signal
Consecutive Loss Streak	Persistent downside pressure

Neon PostgreSQL and SQL Analytics Layer

Cloud Backend

Neon acted as the **analytical database layer**, enabling reusable SQL-based analysis beyond the Python notebook.

✔

These queries made the project reusable, interview-ready, and suitable for real-world analytical exploration.

Reusable SQL Analysis Queries

01	Latest Snapshot of All Coins
02	Daily Returns Time Series — Specific Coin
03	Average Annualized Volatility by Coin
04	Maximum Drawdown by Coin
05	Annualized Return vs Risk Profile
06	Days Above vs Below 200-Day Moving Average
07	BTC vs ETH Daily Return Correlation
08	Sharpe Ratio Approximation per Coin
09	7-Day Rolling Price Change and Momentum Signal
10	Longest Consecutive Loss Streak per Coin

crypto_timeseries

Full daily historical metrics per coin

crypto_snapshot

Latest summary metrics per coin

Apache Superset Hero Dashboard

The dashboard transformed raw market data into an **executive-style analytics interface** for monitoring trend, risk, liquidity, and momentum in one place. Refer image and zip dashboard for the superset dashboard.

\$76.9K

Latest Price

BTC current price KPI

33.31

Ann. Volatility

Annualized volatility KPI

-53.45

Current Drawdown

Peak-to-trough drawdown %

\$615B

Market Cap

BTC market cap KPI



Price + Moving Averages

50-day and 200-day MA overlays



Volatility by Coin

Side-by-side risk comparison



Drawdown Over Time

Persistent stress visualization



Risk-Return Scatter

Multi-asset positioning



Momentum Heatmap

7-day signal visualization

Key Findings from the Analysis



BTC: Highest Scale and Stability

Price ~\$76.9K · Market Cap ~\$615B. Most structurally dominant asset — lowest relative volatility among the three.



SOL: Highest Risk Profile

SOL exhibited the highest annualized volatility, indicating greater upside potential but significantly elevated instability.



Deep Drawdowns Across All Assets

Current drawdown KPI at approximately **-53.45%**. Persistent downside stress observed across major correction phases.



Momentum: Recently Mild Bearish

"Mild Downtrend" signals across BTC, ETH, and SOL in latest readings — short-term weakness despite partial recovery.



Multi-Dimensional Intelligence

Trend, volatility, liquidity, risk-return positioning, and momentum observed in one unified dashboard — more informative than price tracking alone.

What the SQL Layer Added

The SQL layer elevated this project beyond a notebook — enabling **repeatable, database-driven investigation** deployable in any real-world BI environment.

Analytical Capabilities Unlocked

- Compare **risk across assets** using volatility and drawdown queries
- Evaluate **trend regime** using moving-average SQL logic
- Measure **co-movement** via correlation analysis
- Detect **momentum shifts** using LAG and window functions
- Study **persistent loss periods** using streak analysis

Advanced SQL Concepts Demonstrated



CTEs



Window Functions



LAG()



**Rolling
Comparisons**



**Aggregate
Analytics**



CASE Signal Labels

- ✓ This strengthened the project as both a **data engineering** and **business intelligence** portfolio piece.

Recommendations and Next Steps

Business / Analytical

- Use **BTC as baseline benchmark** — largest scale, lowest relative volatility
- Treat **SOL as high-risk** — stricter position sizing and risk controls required
- Combine **momentum + drawdown signals** before directional calls
- Evaluate **risk-adjusted performance**, not raw price growth alone

Technical / Project Improvements



Automate Pipeline

Schedule refreshes with GitHub Actions, Prefect, or Airflow



Expand Universe

Add ADA, XRP, DOGE, BNB for broader coverage



Portfolio Analytics

Add allocation modeling and optimization metrics



Alerting + Demo

Volatility spike alerts and hosted walkthrough for GitHub/LinkedIn



Project Impact

Full Pipeline

API ingestion → computation → cloud storage → BI dashboard in one connected workflow

Tech Stack

Python, SQL, PostgreSQL, and Apache Superset combined in a single production-style project

Financial Depth

Risk metrics, volatility modeling, drawdown analysis, and Sharpe approximation demonstrated

Portfolio-Ready

Aligned with internship and entry-level analytics roles in fintech, consulting, and BI

This project demonstrates the ability to build an end-to-end analytics product — from raw external data ingestion to cloud storage, SQL analysis, and executive dashboarding.

Thank You

*Tools used: Google Colab · Python · pandas · CoinGecko API · Neon
PostgreSQL · Apache Superset · SQL*